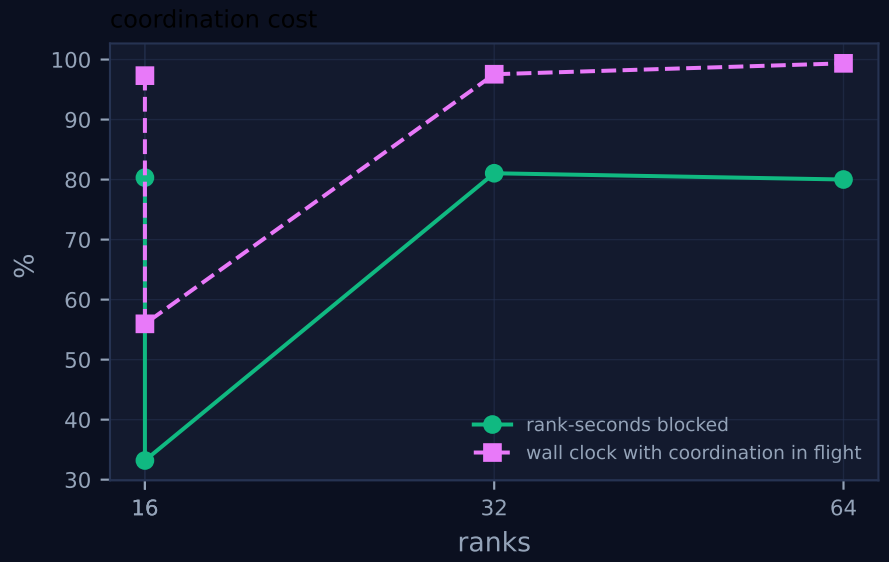
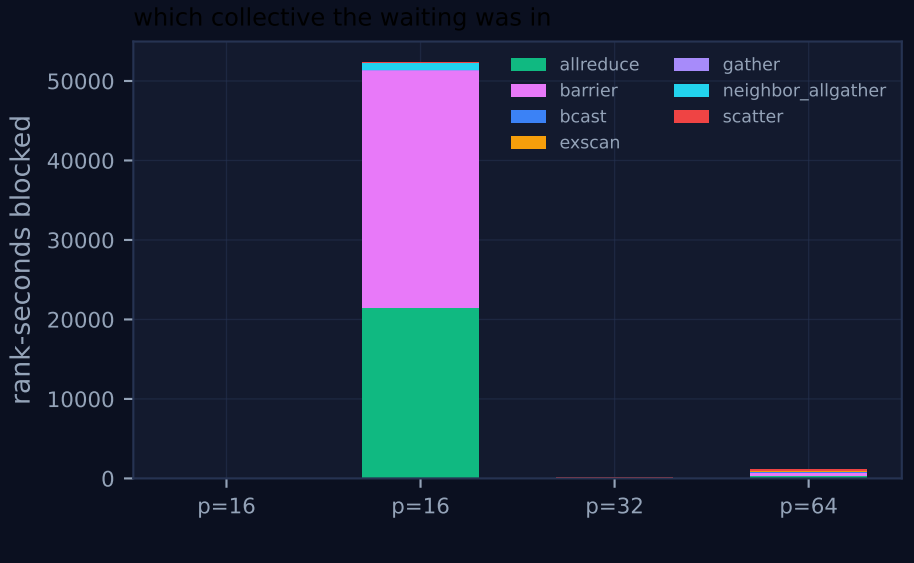

coordination cost
100
90
80
70
60
50
40
30
%
16
32
64
ranks
rank-seconds blocked
wall clock with coordination in flight


disagreement found, not silently resolved
16
14
12
10
8
6
4
2
conflicts lifted
16
32
64
ranks


speedup against one rank
achieved
ideal
2^6
2^5
2^4
2^3
2^2
2^1
2^0
achieved parallelism
16
32
64
ranks


which collective the waiting was in
allreduce
barrier
bcast
exscan
gather
neighbor_allgather
scatter
50000
40000
30000
20000
10000
0
rank-seconds blocked
p=16
p=16
p=32
p=64